



Agitation and Arousal: Pharmacological Management Clinical Guideline in Mental Health Inpatient Settings

Scope

Site	Department, division, or operational area	Applicable to
Royal Perth Bentley Group (RPBG)	Mental Health inpatient areas	Medical, Nursing, Allied Health, Pharmacy Excludes Transitional Care Unit (TCU)* See below

Purpose

To guide clinicians in the pharmacological management of agitation and arousal in admitted patients aged 16 years and above in mental health (MH) settings.

* **Note:** TCU has a sub-acute patient cohort and does not stock or administer intramuscular (IM) agitation and arousal medication and do not currently have the MET equipment to manage an adverse reaction beyond calling 000 and administering oxygen.

Links within document

- [Appendix I: MH pharmacological management of agitation and arousal in Adults \(18-65 years\)](#)
- [Appendix II: MH pharmacological management of agitation and arousal in Older Adults \(65 years+\)](#)
- [Appendix III: MH pharmacological management of agitation and arousal in Youth \(16-17 years\)](#)
- [Appendix IV: MH pharmacological management of agitation and arousal in Pregnant Women](#)
- [Appendix V: Agitation and Arousal PRN Medication Chart example](#)

Procedure

Principles of safe prescribing

- Acute behavioural disturbance reflects an underlying issue or unmet need.¹
- Assessment and management should be multidisciplinary and based on the **Prevention and Management of Aggressive Behaviours within MH Inpatient Setting SOP** and **Safety Planning for Mental Health Consumers Policy**.
- If safe, commence non-pharmacological interventions as the first line option to deescalate the patient and document attempted interventions in the medical notes.
- Consider the flow charts for pharmacological options in specific settings using the following guidelines.
 - For **mild to moderate** risk to patient or others, consider use of oral medications and offer this option to the patient if appropriate, i.e.
 - Agitation and Arousal Score **2-3**
 - For **moderate to severe** risk to patient or others, or if oral medication is not accepted, consider use of IM medication.
 - Agitation and Arousal Score **3-5**
- Consider the following, specific MH pharmacological management of acute behavioural disturbance flowcharts as a 'guideline' only. It is not the intent that the flowcharts take away from clinical decision making processes for individual cases.
 - Adult patients (18-65 years), see [Appendix I](#)
 - Older patients (65 years+), see [Appendix II](#)
 - Youth patients (16-17 years), see [Appendix III](#)
 - Pregnant women, see [Appendix IV](#)
- Treatment options are determined by patient's age and health status (i.e. pregnancy).
- Where the listed treatment options are not suitable, Consultant Psychiatrist input is required to prescribe alternatives and rationale must be documented in the medical record.
- For acute psychosis, including mania, and exacerbation of chronic psychosis, particularly when the patient is hostile or aggressive, refer to **Zuclopenthixol Acetate (Clopixol Acuphase®) Intramuscular Injection SOP**. It is not to be used for rapid tranquilisation or acute sedation.
- Intramuscular lorazepam may be a suitable option for some patients at the discretion of the treating Psychiatrist. Consult your pharmacist if IM lorazepam is required.
- Medications for managing acute agitation and arousal must be prescribed on *WA Agitation and Arousal PRN Medication Chart*.
- The aim of pharmacological management is to reduce the patient's agitation and arousal score to 1 and not to sedate the patient.
- Oral route is preferred over parenteral administration.
- Orally disintegrating tablets, wafers, liquids, or syrups are preferred to tablet to aid compliance and reduce the risk of diversion. Specify the preparation on the medication chart particularly if using liquid or syrups.

Procedure – Principles of safe prescribing cont'd

- Care should be taken with the following subgroups who may require lower doses and increased monitoring for effects of psychotropics (e.g. cardio-respiratory, falls risk) using *Adult Observation and Response Chart (AORC)*. START LOW, GO SLOW and GET HELP
 - Neuroleptic naïve
 - Dehydrated or renal failure
 - At the extremes of weight range (e.g. Underweight)
 - Obstructive sleep apnoea
 - Experiencing substance intoxication or withdrawal or known legal and illegal substances recently taken
 - Multiple co-morbidities
 - Pregnant
 - Epileptic
 - Under 24 years
 - Sensitive to extra-pyramidal side effects (EPSE)
 - Frail persons - exercise clinical judgement / caution, otherwise refer to the [Clinical Frailty Scale](#)² for guidance. Consider a score of 5 as 'frail' and use of frail dosages.
 - Culturally diverse – Persons of Aboriginal and Torres Strait Islander descent and those of Southeast Asian and African descent may have genetic differences in cytochrome P450 and 2D6 metabolism causing predisposition to increased toxicities and side effects³
 - Patients who have specific dietary requirements, e.g., vegans, people of certain faith. Olanzapine wafers contain Bovine Gelatine which may not be appropriate.
- Consider **capacity and consent** deliberations and ensure these are documented in the medical record. Refer to **WA Department of Health Consent to Treatment Policy**. If patient lacks capacity and / or is at imminent risk of harm to self or others, the Resident Medical Officer (RMO) or senior nursing staff to determine if restraint or seclusion or pharmacological management of patient may be required/review for **Mental Health Act (MHA) 2014** applications and requirements, e.g., 6A or 6B Form, 9A - Record of Emergency Psychiatric Treatment and/or the doctrine of necessity.
- Consideration should be given to any known pre-existing medical illnesses, drug allergies and any regularly prescribed oral or depot medication, as this may impact on dose requirements and potential side effects.
- An exact dose must be specified, not a dosage range.
- Each medication order must be for **one** route only. The 'Max dose/24 hr' prompt on the *WA Agitation and Arousal PRN Medication Chart* indicates the total amount of the medication, including both oral and IM formulations, which may be administered for PRN doses only for that medication. The maximum dose allowed within a 24-hour time period does not relate to the calendar day. To determine whether a patient can have a subsequent dose, calculate the cumulative dose of the PRN medication received in the preceding 24 hours, and ensure a subsequent dose will not exceed the maximum specified on the PRN order.

Procedure – Principles of safe prescribing cont'd

- The prescriber should ensure that the maximum recommended daily dose of a prescribed medication on the PRN order is not exceeded by taking into account any regular doses of that medication prescribed in the 'regular medicines' section of the relevant *WA Hospital Medication Chart (HMC)*.
- Antipsychotic polypharmacy (concurrent use of two or more antipsychotics, including those prescribed regularly) and high dose prescribing (doses in excess of the licensed maximum daily dose as outlined in the Australian Product Information leaflet) carries risks, therefore the use of antipsychotics for the management of agitation and arousal should be monitored for short term and cumulative adverse effects.
- Unless contraindicated, benztropine should be prescribed in the 'when required' (PRN) section of the *WA HMC* when typical antipsychotics are prescribed especially haloperidol. The usual dose of benztropine is oral (PO) or IM 1mg – 2mg TDS PRN.
- Baseline 12 lead electrocardiogram (ECG) monitoring is strongly recommended before prescribing and administering parenteral antipsychotics. ECG monitoring is recommended for all patients receiving antipsychotic affecting QTC interval in any formulation based on clinical need and presentation. If it is not possible to perform an ECG, the reason must be clearly documented in the patient medical record.
- The rationale for any changes or deviations from this guideline should be clearly documented in the medical notes.
- The treating team or Duty Medical Officer (DMO) should be notified if an IM injection has been given for rapid tranquilisation. A doctor should aim to be at the scene within 30 minutes. Medical support must be available in case of adverse reactions, over sedation (RR< 10 breaths per minute) or the need to administer flumazenil (to reverse sedation, drowsiness). Contact the DMO or RMO (if DMO not on-shift) or initiate code blue.
 - Resuscitation equipment and medication must be available and easily accessible, and staff should be familiar with their use. The equipment must be maintained and checked as per the **Medical Emergencies and Cardiorespiratory Arrest Management CPS**
 - Access to a fully equipped resuscitation trolley and Medical Emergency Team (MET) including intravenous fluids and administration sets.
 - Access to an ECG machine.
 - In case a patient develops severe EPSE, all wards should have benztropine injection.

Procedure cont'd

Agitation and arousal scores

- Agitation and arousal scores must be documented on the *WA Agitation and Arousal PRN Medication Chart* (see table 1). See completed example: [Appendix V: Agitation and Arousal PRN Medication Chart example](#)

Table 1: Agitation and arousal rating scale and associated management

Score	Patient's state	Management
0	Asleep or unconscious	Review prescribed doses and conduct required monitoring.
1	Settled, minimal agitation	Optimal state: PRN medication not required.
2	Mildly aroused, pacing, still willing to talk reasonably	1. Attempt to verbally de – escalate the patient, as appropriate 2. Offer oral treatment relevant to patient's current rating. 3. IM option only if patient refuses oral treatment.
3	Moderately aroused, agitated, becoming more vocal and unreasonable or hostile	
4	Highly aroused and possibly distressed or fearful, possibly noisy but not violent	
5	Highly aroused and violent toward self, others and/or property	

- Agitation and arousal scores must be documented on the *WA Agitation and Arousal PRN Medication Chart*:
 - Before each PRN dose, and
 - Reviewed and documented after:
 - 60 and 90 minutes for all oral doses
 - 30 and 60 minutes for all IM doses.

Repeating doses

- Where additional doses are required, administer at the frequency indicated until a score of 1 or maximum daily dosage is achieved:
 - Oral: May be repeated hourly (except in pregnancy, refer to Appendix IV)
 - IM: May be repeated after 30 minutes **except** olanzapine IM and clonazepam IM.
 - Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose if required. Total of 3 doses or 30mg maximum in 24 hours (10mg maximum in 24 hours for older adults over 65 years) whichever occurs first.
 - Clonazepam IM may be repeated after 3 hours, maximum 4mg in 24 hours (higher doses may be used in exceptional circumstances after discussion with Consultant Psychiatrist).
- Dosing frequencies outside the recommended guidelines require Consultant Psychiatrist approval.
- Seek the advice of a Consultant Psychiatrist if not achieving a score of 1 at maximum doses recommended.

Procedure cont'd

Monitoring

- Following the administration of medication for agitation and arousal
 - If safe, monitor the patient's vital signs and record on the *AORC* and *WA Agitation and Arousal PRN Medication Chart* to assess efficacy of treatment, required escalation and document in the inpatient notes.
 - All physiological observations must be taken to allow monitoring of any physical deterioration and recorded on the *AORC*. The observations must include:
 - Pulse
 - Blood pressure (BP)
 - Temperature
 - Oxygen saturation
 - Consciousness levels
 - Respiratory rate
 - Observe for adverse effects: monitor for signs of EPSE (and in particular, laryngeal dystonia), seizures, breathing difficulties, neuroleptic malignant syndrome, following the administration of antipsychotics.
- Frequency of monitoring:
 - Medical staff should advise and document in the inpatient notes' medications used, monitoring and observation requirements as per the *WA Agitation and Arousal PRN Medication Chart* and/or any additional monitoring, if required.

Table 2 Frequency of physiological observations post PO and IM medication

Post PO medication	Review the patient after ONE hour. Further monitoring beyond 1 hour should be considered if deemed clinically appropriate and record arousal score on Agitation and Arousal Score.
Post IM medication	1st hour: every 15 minutes Thereafter: every 30 minutes until patient is ambulating or minimum 4 hours post IM dose and record arousal score on Agitation and Arousal Score.

- Scores should be monitored and responses as per instructions on the *Adult Deterioration Detection System (ADDS) Score Chart*.
- Seek urgent medical advice or initiate code blue protocol if the patient is unconscious or if there is a sudden deterioration in any of the basic observations of temperature, pulse, BP, or oxygen saturation. Refer to **Recognising and Responding to Acute Deterioration (RRAD) Policy** and **RRAD in Mental State, Cognition and Behaviour SOP**
- If all physiological observations cannot be carried out, or the patient refuses, non-contact observations must still be completed. Respiratory rate and level of consciousness using the Alert, Voice, Pain, Unresponsive (AVPU) system must be recorded. This would include patients held in seclusion. Nursing staff must annotate "®" for refused for other vitals on *AORC* and this detail must be documented in the medical record. Observations should be re-offered or re-attempted as the situation dictates.

Procedure – Monitoring cont'd

- If the patient is asleep, the use of pulse oximetry to measure oxygen saturation (SpO₂) is recommended until patient is ambulatory every 4 hours.
 - Patients with abnormal vital signs (SpO₂ and respiratory rate) should be reassessed a minimum of every 2 hrs for the first 4 hrs, then every 4 hrs if clinically stable
- 12 lead ECG should be performed post parenteral sedation and reviewed for signs of conduction disturbance and to assess for prolongation of QTc interval. Haematological monitoring should be performed at the earliest opportunity when parenteral antipsychotics are given. Hypokalaemia, stress and agitation place the patient at risk of cardiac arrhythmias.

Medication review

- The agitation and arousal treatment regimen should be reviewed in conjunction with the patient's regularly prescribed medications. Frequent doses of PRN medications may indicate a need to modify the patient's regular regimen.
- The treating team Medical Officer must review the *WA Agitation and Arousal PRN Medication Chart* every 24 hours if any of the steps on the chart are utilised during weekdays, as applicable. During out of hours, weekends and public holidays the chart should be reviewed when requested by a nurse or if 2 or more steps had to be utilised to reduce arousal. This review must be documented in the medical record and as per the **Minimum Requirement for Medical Review of Consumers Admitted to Inpatient MH Wards or CMHS Policy**.
- Also refer to **Escalation to Consultants Policy**.

Handover

- Use iSoBAR format. Refer to **Clinical Handover Policy**
- To senior clinician(s)
- To treating team
- To Safety for Everyone (SAFE)/BeSAFE (Bentley Health Service) as needed.

Roles and responsibilities

Role	Responsibility
Medical Officer	To ensure prescribing complies with this document (and rationale is articulated when it does not), management plans are formulated and reviewed, with clear documentation of these processes in the medical record.
Pharmacist	To review the medication charts and address any concerns with the prescriber, including escalation to the team Consultant Psychiatrist, or to the Clinical Lead Inpatient or MH Medical Co-Director when required.
Nurse	To ensure that the required observations are checked and recorded on <i>AORC</i> and to escalate any concerns as appropriate.

Compliance monitoring

Compliance with this Guideline will be monitored via the following:

- Datix CIMS and/or CHOIR notifications
- Compliments and Complaints.

Facilitator

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Authors / acknowledgements

We acknowledge the following current site endorsed work and/or contributors used to compile this document.

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Armadale Kalamunda Group. (2021). Agitation and Arousal: Pharmacological Management Guideline. Western Australia.

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RPBG Medication Safety Committee (13/06//2024)

Related policies, practice standards, clinical guidelines

RPBG Policy Hub

- [Medication Preparation and Routes of Administration CPS](#)
- [Medication Management and Storage Clinical Guideline](#)
- [Vital Signs for Adults NPS](#)
- [Special and Supportive Observations Policy](#)
- [Comprehensive Care SOP](#)
- [Minimisation and Safe Management of Restraint and Seclusion \(for Mental Health Areas\) CPS](#)
- [SAFE and BeSAFE SOP](#)
- [Prevention and Management of Aggressive Behaviours within MH Inpatient Setting SOP](#)
- [Zuclopenthixol Acetate \(Clopixol Acuphase®\) Intramuscular Injection SOP](#)
- [Medical Emergencies and Cardiorespiratory Arrest Management CPS](#)
- [Recognising and Responding to Acute Deterioration \(RRAD\) Policy](#)
- [RRAD in Mental State, Cognition and Behaviour SOP](#)
- [Minimum Requirement for Medical Review of Consumers Admitted to Inpatient MH Wards or CMHS Policy](#)
- [Clinical Handover Policy](#)
- [Escalation to Consultants Policy](#)

EMHS Policy Hub

- [Consent Policy](#)

WA Department of Health Policy Hub

- [Medication Chart Policy](#)
- [Medication Review Policy](#)
- [Statewide Standardised Clinical Documentation for Mental Health Services Policy](#)
- [Safety Planning Procedures for Mental Health Consumers](#)
- [Principles and Best Practice for the Care of People Who May Be at Risk of Exhibiting Violent or Aggressive Behaviour](#)
- [Principles and Best Practice for the Care of People Who May Be Suicidal](#)
- [Consent to Treatment Policy](#)

Other

- [WA Mental Health Act \(MHA\) 2014](#)
- [WA Medicines and Poisons Act 2014](#)
- [WA Medicines and Poisons Regulations 2016](#)
- [WA Work Health and Safety Act 2020](#)
- [WA MHA 2014 Charter of Mental Health Care Principles](#)
- [WA Department of Health Medication Information Printable Consumer Leaflets](#)
- [Chief Psychiatrist's Standards for Clinical Care](#)

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Authorised by:	Deliver What Matters Most Committee 24/09/2024	Revision Due:	09/2026

Related forms

- *WA Agitation and Arousal PRN Medication Chart*
- *Adult Observation and Response Chart (AORC)*
- *WA Hospital Medication Chart (WA HMC)*
- *Adult Deterioration Detection System (ADDS) Score Chart*

Legislative requirements, the evidence, and the standard

For information relating to the legislative requirements and standards that RPBG policy documents must adhere to, and regarding the logos and levels of evidence used within RPBG policy documents, refer to [Legislative Requirements, the Evidence and the Standard](#) (live link) on the Policy Hub.

Related national standards

ACSQHC NSQHS Standards 2nd Edition (2021): Standard 1: Clinical Governance; Standard 2: Partnering with Consumers; Standard 4: Medication Safety; Standard 5: Comprehensive Care; Standard 6: Communicating for Safety; Standard 8: Recognising and Responding to Acute Deterioration

NSMHS Standards (2010): Standard 1: Rights and responsibilities; Standard 2: Safety; Standard 8 - Governance, leadership and management; Standard 10: Delivery of care

References

1. Therapeutic Guidelines Ltd. (March 2021). *Principles of care for a patient with acute behavioural disturbance*. [Therapeutic Guidelines \(health.wa.gov.au\)](http://online.tg.org.au.smhslibresources.health.wa.gov.au/ip/desktop/index.htm)
<http://online.tg.org.au.smhslibresources.health.wa.gov.au/ip/desktop/index.htm>
2. Dalhousie University. *Clinical Frailty Scale*. Geriatric Medicine Research. Canada. [Clinical Frailty Scale - Geriatric Medicine Research - Dalhousie University](#)
3. Tornio A, Backman JT. Cytochrome P450 in Pharmacogenetics: An Update. *Adv Pharmacol*. 2018;83:3-32. doi: 10.1016/bs.apha.2018.04.007. PMID: 29801580.
4. NICE (National Institute for Health and Care Excellence). (2015). *Violence and aggression: short term management in mental health, health and community settings*. UK. <https://www.nice.org.uk/guidance/ng10>
5. Taylor DM, Barnes TRE, Young AH. (2021). *The Maudsley – Prescribing Guidelines in Psychiatry 14th Edition*. Wiley Blackwell.

Appendix I: MH pharmacological management for arousal and agitation in Adults (18-65 years)

STEP 1 - De-escalation techniques will be attempted first for mildly to moderately aroused patients before progressing to psychotropic medications															
ORAL		IM (**ECG Highly recommended)													
STEPS 2 & 3 - MILD TO MODERATE AROUSAL: MH Agitation and arousal scores 2 - 3 and willing to take oral medication *Specify EXACT DOSE, DO NOT use Dosing ranges		STEP 4 - MODERATE TO SEVERE AROUSAL. MH Agitation and arousal scores 3 - 5 and/or unwilling to take oral medication *Specify EXACT DOSE, DO NOT use Dosing ranges													
First Choice Lorazepam - Adult: 1-2 mg PO (max dose 6 mg in 24 hours) - Frail: 0.5-1 mg PO (max dose 2 mg in 24 hours) Second Choice Quetiapine IR (Immediate Released) - Adult: 25-100 mg PO (max dose 1000 mg in 24 hours) - Frail: 12.5-25 mg PO (max dose 100 mg in 24 hours) Third Choice Olanzapine - Adult: 5-10 mg PO (max dose 30 mg in 24 hours) - Frail: 2.5- 5 mg PO (max dose 10 mg in 24 hours)		First Choice Olanzapine - Adult: 10 mg IM (max dose 30 mg in 24 hours) - Frail: 2.5-5 mg IM (max dose 10 mg in 24 hours) Do not administer IM olanzapine within 2 hours before or after any parenteral benzodiazepine Second Choice Clonazepam - Adult: 1-2 mg IM (max dose 4 mg in 24 hours) - Frail: 1mg IM (max dose 2 mg in 24 hours) Third Choice Haloperidol - Adult: 0.5-10 mg IM (max dose 20 mg in 24 hours) - Frail: 0.25-0.5 mg IM (max dose 2 mg in 24 hours)													
MONITORING • Observations including but not limited to blood pressure and respiratory function and recorded on AORC as per Vital Signs NPS • Observations completed until the patient is ambulatory or at least 4 hours post administration of IM medications • Be aware of potential for QTc prolongation, acute dystonia, and Neuroleptic Malignant Syndrome with antipsychotics • ECG post IM antipsychotics where possible • Frequent PRN administrations for more than 24 hours should prompt review of regular medication regimen • Medical staff to review for MHA 2014 applications and requirements		REVIEW RESPONSE • Review all IM doses after 30 and 60 minutes and record Agitation & Arousal score • Review all oral doses after 60 and 90 minutes and record Agitation & Arousal score REPEAT DOSING • Oral: May be repeated after 30 minutes • IM: May be repeated after at least 30 minutes except for • IM Olanzapine may be repeated after 2 hours, and a third dose may be given 6 hours after the first. Total of 3 doses/30 mg/24 hours (or 3 doses/10 mg/24 hours for frail or older adults over 65 years) • IM Clonazepam may be repeated after 3 hours, maximum 4 mg in 24 hours • Seek the advice of a senior clinician if not achieving a score of 1 at maximum recommended doses.													
		MEDICATION SELECTION/DOSES • Where possible use PRN medications that are the same/like regular medications • Dosing frequencies outside the recommended time and/or dose range require senior clinician input OTHER MEDICATIONS • Benzatropine (oral or IM) 1 mg to 2 mg 'TDS PRN' should be prescribed on the PRN section of the <i>WA Hospital Medication Chart</i> NOTIFICATIONS • After an IM dose is administered the doctor on duty should be notified													
		**Recommended Maximum Doses (Per 24 hours) <table><tr><th>Medication</th><th>TOTAL Daily Dosing prn + Regular</th></tr><tr><td>Lorazepam PO</td><td>6 mg (frail 2 mg)</td></tr><tr><td>Quetiapine IR PO</td><td>1000 mg (frail 100 mg)</td></tr><tr><td>Olanzapine PO/IM</td><td>30 mg (frail 10 mg)</td></tr><tr><td>Clonazepam PO/IM</td><td>4 mg (frail 2 mg)</td></tr><tr><td>Haloperidol PO/IM</td><td>20 mg (frail 2 mg)</td></tr></table>		Medication	TOTAL Daily Dosing prn + Regular	Lorazepam PO	6 mg (frail 2 mg)	Quetiapine IR PO	1000 mg (frail 100 mg)	Olanzapine PO/IM	30 mg (frail 10 mg)	Clonazepam PO/IM	4 mg (frail 2 mg)	Haloperidol PO/IM	20 mg (frail 2 mg)
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Appendix II: MH pharmacological management for arousal and agitation in Older Patients (over 65 years)

STEP 1 - De-escalation techniques will be attempted first for mildly to moderately aroused patients before progressing to psychotropic medications																		
If the patient has Parkinsonism, use only Quetiapine 12.5mg (frail patients) 4 hourly pm: Maximum 100mg in 24 hours- Do not proceed onto Steps 2 - 4																		
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Equally preferential Haloperidol - Adult: 0.25- 0.5 mg PO (max dose 2 mg in 24 hours) OR Risperidone - Adult: 0.25 -0.5 mg PO (max dose 2 mg in 24 hours) OR Quetiapine IR (Immediate Released) - Adult: 12.5 -25 mg PO (max dose 100 mg in 24 hours) Upon Senior Review, proceed to Step 4		Equally preferential Haloperidol Adult: 0.25- 0.5 mg IM (max dose 2 mg in 24 hours) OR Olanzapine - Adult: 2.5 mg PO or IM (max dose 10 mg in 24 hours) Do not administer IM olanzapine within 2 hours before or after any parental benzodiazepine																
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Appendix III: MH pharmacological management for arousal and agitation in Youth (16-17 years)

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First Choice Lorazepam Young Adult: 0.5 - 2 mg PO (max dose 6 mg in 24 hours) Second Choice Quetiapine IR (Immediate Released) 40 kg and over: 25 – 50 mg PO (max dose 750 mg in 24 hours) Less than 40 kg: 12.5 – 25 mg PO (max dose 200 mg in 24 hours) Third Choice Olanzapine 40 kg and over: 5 – 10 mg PO (max dose 30 mg in 24 hours) Less than 40 kg: 2.5 – 10 mg PO (max dose 10 mg in 24 hours)		First Choice Olanzapine 40 kg and over: 5 – 10 mg IM (max dose 30 mg in 24 hours) Less than 40 kg: 2.5 – 10 mg IM (max dose 10 mg in 24 hours) Do not administer IM olanzapine within 2 hours before or after any parental benzodiazepine Second Choice Clonazepam Young Adult: 1 – 2 mg IM (max dose 4 mg in 24 hours) OR Midazolam Young Adult: 2.5– 5 mg IM (max dose 20 mg in 24 hours)																			
MONITORING - Observations including but not limited to blood pressure and respiratory function and recorded on AORC as per Vital Signs NPS - Observations completed until the patient is ambulatory or at least 4 hours post administration of IM medications - Be aware of potential for QTc prolongation, acute dystonia, and Neuroleptic Malignant Syndrome with antipsychotics - ECG post IM antipsychotics where possible - Frequent PRN administrations for more than 24 hours should prompt review of regular medication regimen - Medical staff to review for MHA 2014 applications and requirements	REVIEW RESPONSE - Review all IM doses after 30 and 60 minutes and record Agitation & Arousal score - Review all oral doses after 60 and 90 minutes and record Agitation & Arousal score REPEAT DOSING - Oral: May be repeated after 30 to 60 minutes and repeated as necessary as per Child & Adolescent Health Service - Arousal and Agitation Management for Mental Health Inpatients Guideline - IM: May be repeated after at least 30 minutes except for - IM Olanzapine may be repeated after 2 hours and a third dose may be given 6 hours after the first. Total of 3 doses/30 mg/24 hours for patient 40 kg and over (or 3 doses/10 mg/24 hours for patient less than 40 kg). - IM Clonazepam may be repeated after 3 hours, maximum 4 mg in 24 hours - Seek the advice of a senior clinician if not achieving a score of 1 at maximum recommended doses.	MEDICATION SELECTION/DOSES - Where possible use PRN medications that are the same/like regular medications - Dosing frequencies outside the recommended time and/or dose range require senior clinician input OTHER MEDICATIONS Ensure Flumazenil is available and charted if benzodiazepines are administered parenterally - Benzatropine (oral or IM 1 mg to 2 mg 'TDS PRN' should be prescribed on the PRN section of the <i>WA Hospital Medication Chart</i> NOTIFICATIONS - After an IM dose is administered the doctor on duty should be notified	**Recommended Maximum Doses (Per 24 hours) <table><tr><th>Medication</th><th>Recommended Maximum prn</th><th>TOTAL Daily Dosing prn + Regular</th></tr><tr><td>Quetiapine IR PO</td><td>12.5-25 mg (less than 40 kg) 25-50 mg (40 kg and over)</td><td>200 mg (less than 40 kg) 750 mg (40 kg and over)</td></tr><tr><td>Lorazepam PO</td><td>0.5-2 mg</td><td>6 mg</td></tr><tr><td>Olanzapine PO/IM</td><td>5 mg (less than 40 kg) 10 mg (40 kg and over)</td><td>10 mg (less than 40 kg) 30 mg (40 kg and over)</td></tr><tr><td>Clonazepam IM</td><td>1-2 mg</td><td>4 mg</td></tr><tr><td>Midazolam IM</td><td>5 mg</td><td>20 mg</td></tr></table>	Medication	Recommended Maximum prn	TOTAL Daily Dosing prn + Regular	Quetiapine IR PO	12.5-25 mg (less than 40 kg) 25-50 mg (40 kg and over)	200 mg (less than 40 kg) 750 mg (40 kg and over)	Lorazepam PO	0.5-2 mg	6 mg	Olanzapine PO/IM	5 mg (less than 40 kg) 10 mg (40 kg and over)	10 mg (less than 40 kg) 30 mg (40 kg and over)	Clonazepam IM	1-2 mg	4 mg	Midazolam IM	5 mg	20 mg
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Agitation and Arousal: Pharmacological Management Clinical Guideline in Mental Health Inpatient Settings

Version: 1.0

Endorsed by: Drugs and Therapeutics Committee 06/08/2024

Authorised by: Deliver What Matters Most Committee 24/09/2024

Date Compiled: 04/2024

Executive Sponsor: Medical Co-Director, Mental Health Division

Revision Due: 09/2026

Appendix IV: Mental Health pharmacological management for arousal and agitation in Pregnant Women

STEP 1 - De-escalation techniques will be attempted first for mildly to moderately aroused patients before progressing to psychotropic medications

ORAL	IM (**ECG Highly recommended)
STEPS 2 & 3 - MILD TO MODERATE AROUSAL: MH Agitation and arousal scores 2 - 3 and willing to take oral medication *Specify EXACT DOSE, DO NOT use Dosing ranges	STEP 4 - MODERATE TO SEVERE AROUSAL. MH Agitation and arousal scores 3 -5 and/or unwilling to take oral medication *Specify EXACT DOSE, DO NOT use Dosing ranges
First Choice Quetiapine IR (Immediate Released) - Pregnant: 12.5 -50 mg PO (max dose 400 mg in 24 hours)	First Choice Olanzapine - Pregnant: 10 mg IM (max dose 30 mg in 24 hours)
Second Choice Lorazepam - Pregnant: 1-2 mg PO (max dose 4 mg in 24 hours)	Second Choice Droperidol - Pregnant: 10 mg IM (max dose 20 mg in 24 hours)
Third Choice Olanzapine - Pregnant: 5-10 mg PO (max dose 30 mg in 24 hours)	Midazolam - Pregnant: 5-10 mg IM (max dose 10 mg in 24 hours)

MEDICATION SELECTION/DOSES - Where possible use PRN medications that are the same/like regular medications • Dosing frequencies outside the recommended time and/or dose range require senior clinician input MONITORING - Observations including but not limited to blood pressure and respiratory function and recorded on AORC as per Vital Signs NPS - Observations completed until the patient is ambulatory or at least 4 hours post administration of IM medications - Be aware of potential for QTc prolongation, acute dystonia, and Neuroleptic Malignant Syndrome with antipsychotics - ECG post IM antipsychotics where possible - Frequent PRN administrations for more than 24 hours should prompt review of regular medication regimen - Medical staff to review for MHA 2014 applications and requirements NOTIFICATIONS - After an IM dose is administered the doctor on duty should be notified	REVIEW RESPONSE - Review all IM doses after 30 and 60 minutes and record Agitation & Arousal score - Review all oral doses after 60 and 90 minutes and record Agitation & Arousal score REPEAT DOSING - Oral: May be repeated after 2 to 6 hourly - IM: May be repeated after at least 30 minutes except for IM Olanzapine may be repeated after 2 hours and a third dose may be given 6 hours after the first. Total of 3 doses/30mg/24 hours - Seek the advice of a senior clinician if not achieving a score of 1 at maximum recommended doses. OTHER MEDICATIONS - Ensure Flumazenil is available and charted if benzodiazepines are administered parenterally. - Benzatropine (oral or IM 1 mg to 2 mg 'TDS PRN' is NOT routinely advised. Only used if EPSE is assessed and must be prescribed on the PRN section of the <i>WA Hospital Medication Chart</i>	**Recommended Maximum Doses (Per 24 hours) <table><tr><th>Medication</th><th>TOTAL Daily Dosing prn + Regular</th></tr><tr><td>Quetiapine IR PO</td><td>400 mg</td></tr><tr><td>Lorazepam PO</td><td>4 mg</td></tr><tr><td>Midazolam IM</td><td>10 mg</td></tr><tr><td>Olanzapine PO/IM</td><td>30 mg</td></tr><tr><td>Droperidol</td><td>20 mg</td></tr></table>	Medication	TOTAL Daily Dosing prn + Regular	Quetiapine IR PO	400 mg	Lorazepam PO	4 mg	Midazolam IM	10 mg	Olanzapine PO/IM	30 mg	Droperidol	20 mg	***Medication Details <table><tr><th>Drug</th><th>Time (Min to Onset)</th><th>Peak Effect (Min)</th><th>Half-Life (hr)</th></tr><tr><td>Quetiapine IR PO</td><td>30</td><td>90</td><td>46 hrs</td></tr><tr><td>Lorazepam PO</td><td>30</td><td>90</td><td>8 hrs</td></tr><tr><td>Olanzapine PO</td><td>60</td><td>360 (6 hrs)</td><td>30 hrs</td></tr><tr><td>Droperidol IM</td><td>10</td><td>30</td><td>4 hrs</td></tr><tr><td>Olanzapine IM</td><td>15</td><td>45</td><td>30 hrs</td></tr><tr><td>Midazolam IM</td><td>15</td><td>60</td><td>2 hrs</td></tr></table>	Drug	Time (Min to Onset)	Peak Effect (Min)	Half-Life (hr)	Quetiapine IR PO	30	90	46 hrs	Lorazepam PO	30	90	8 hrs	Olanzapine PO	60	360 (6 hrs)	30 hrs	Droperidol IM	10	30	4 hrs	Olanzapine IM	15	45	30 hrs	Midazolam IM	15	60	2 hrs
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